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Phase continuity handling for a UE CSI report of time domain channel properties measurements

Abstract

Systems and methods are provided for to support user equipment (UE) channel state information (CSI) report of time domain channel properties. Embodiments of a UE herein may receive, from a network node, a CSI report configuration (CSI-ReportConfig). The CSI-ReportConfig may comprise details regarding measurement and reporting of time domain channel properties (TDCP) based on a CSI reference signal (CSI-RS) for tracking reference signal (TRS). The UE may determine invalid CSI-RS resources and valid CSI-RS resources based on a timing of a duplex direction change. The UE may measure and report TDCP based on the valid CSI-RS resources.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2016/0241434	12/2015	Li	N/A	H04L 41/0803
2018/0242164	12/2017	Lee	N/A	H04L 1/0026
2018/0367289	12/2017	Kim	N/A	H04W 72/0453
2019/0281487	12/2018	Liu et al.	N/A	N/A
2020/0084714	12/2019	Medles et al.	N/A	N/A
2021/0400677	12/2020	Lee et al.	N/A	N/A
2022/0256458	12/2021	Noh et al.	N/A	N/A
2022/0279518	12/2021	Lin et al.	N/A	N/A
2022/0417776	12/2021	Manolakos et al.	N/A	N/A
2023/0246687	12/2022	Rahman et al.	N/A	N/A
2023/0254718	12/2022	Rahman et al.	N/A	N/A
2023/0370139	12/2022	Rahman et al.	N/A	N/A
2023/0422073	12/2022	Liu et al.	N/A	N/A
2024/0421877	12/2023	Frenne	N/A	H04B 7/046

OTHER PUBLICATIONS

Interdigital, Inc. , “Aspects of CSI Enhancements”, R1-2203380, 3GPP TSG RAN WG1 #109-e, e-Meeting, Agenda Item 9.1.2, May 9-20, 2022, 7 pages. cited by applicant

Nokia , “Corrections on eIAB”, R1-2205677, Change Request 38.214 CR 0286, Current Version 17.1.0, May 9-20, 2022, 4 pages. cited by applicant

PCT/US2023/071907 , International Search Report and Written Opinion, Nov. 28, 2023, 17 pages. cited by applicant

Samsung , “NR MIMO evolution for downlink and uplink”, RP-221393, 3GPP TSG RAN Meeting #96, Budapest, Hungary, Agenda Item 9.3.1.4, Jun. 6-9, 2022, 21 pages. cited by applicant

ZTE , “CSI enhancement for high/medium UE velocities and CJT”, R1-2203265, 3GPP TSG RAN WG1 Meeting #109-e, e-Meeting, Agenda Item 9.1.2, May 9-20, 2022, 14 pages. cited by applicant

U.S. Appl. No. 17/935,000, Non-Final Office Action, Jan. 3, 2025, 16 pages. cited by applicant

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Background/Summary

TECHNICAL FIELD

(1) This application relates generally to wireless communication systems, including reporting of time domain channel properties.

BACKGROUND

(2) Wireless mobile communication technology uses various standards and protocols to transmit data between a base station and a wireless communication device. Wireless communication system standards and protocols can include, for example, 3rd Generation Partnership Project (3GPP) long term evolution (LTE) (e.g., 4G), 3GPP new radio (NR) (e.g., 5G), and IEEE 802.11 standard for wireless local area networks (WLAN) (commonly known to industry groups as Wi-Fi®).

(3) As contemplated by the 3GPP, different wireless communication systems standards and protocols can use various radio access networks (RANs) for communicating between a base station of the RAN (which may also sometimes be referred to generally as a RAN node, a network node, or simply a node) and a wireless communication device known as a user equipment (UE). 3GPP RANs can include, for example, global system for mobile communications (GSM), enhanced data rates for GSM evolution (EDGE) RAN (GERAN), Universal Terrestrial Radio Access Network (UTRAN), Evolved Universal Terrestrial Radio Access Network (E-UTRAN), and/or Next-Generation Radio Access Network (NG-RAN).

(4) Each RAN may use one or more radio access technologies (RATs) to perform communication between the base station and the UE. For example, the GERAN implements GSM and/or EDGE RAT, the UTRAN implements universal mobile telecommunication system (UMTS) RAT or other 3GPP RAT, the E-UTRAN implements LTE RAT (sometimes simply referred to as LTE), and NG-RAN implements NR RAT (sometimes referred to herein as 5G RAT, 5G NR RAT, or simply NR). In certain deployments, the E-UTRAN may also implement NR RAT. In certain deployments, NG-RAN may also implement LTE RAT.

(5) A base station used by a RAN may correspond to that RAN. One example of an E-UTRAN base station is an Evolved Universal Terrestrial Radio Access Network (E-UTRAN) Node B (also commonly denoted as evolved Node B, enhanced Node B, eNodeB, or eNB). One example of an NG-RAN base station is a next generation Node B (also sometimes referred to as a g Node B or gNB).

(6) A RAN provides its communication services with external entities through its connection to a core network (CN). For example, E-UTRAN may utilize an Evolved Packet Core (EPC), while NG-RAN may utilize a 5G Core Network (5GC).

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) To easily identify the discussion of any particular element or act, the most significant digit or

digits in a reference number refer to the figure number in which that element is first introduced.

(2) FIG. 1 illustrates a signal flow diagram for supporting UE CSI report of time domain channel properties in accordance with some embodiments.

(3) FIG. 2 illustrates a UCI assembly in accordance with some embodiments.

(4) FIG. 3 illustrates a scheduling restriction based on the DRX active time in accordance with some embodiments.

(5) FIG. 4 illustrates a scheduling restriction based on a number of slots in accordance with some embodiments.

(6) FIG. 5 illustrates a method for a UE to handle phase continuity in accordance with some embodiments.

(7) FIG. 6 illustrates a timeline of a phase continuity interruption in accordance with some embodiments.

(8) FIG. 7 illustrates a timeline of a second case of phase continuity interruption in accordance with some embodiments.

(9) FIG. 8 illustrates an example architecture of a wireless communication system, according to embodiments disclosed herein.

(10) FIG. 9 illustrates a system for performing signaling between a wireless device and a network device, according to embodiments disclosed herein.

DETAILED DESCRIPTION

(11) Various embodiments are described with regard to a user equipment (UE). However, reference to a UE is merely provided for illustrative purposes. The example embodiments may be utilized with any electronic component that may establish a connection to a network and is configured with the hardware, software, and/or firmware to exchange information and data with the network.

Therefore, the UE as described herein is used to represent any appropriate electronic component.

(12) Certain systems (e.g., 3GPP Release (Rel) 15, 16, and 17) support advanced channel state information (CSI) reporting by exploiting channel correlations. For example, channel spatial correlation can be exploited to support higher resolution CSI feedback in both the Type I and Type II multiple input multiple output (MIMO) codebook. Channel frequency correlation can be exploited to reduce CSI overhead in Type II CSI Rel-16 enhancement. Channel downlink (DL) and uplink (UL) correlation can be exploited for reciprocity based MIMO. For UL operation, systems can support non-codebook based Physical Uplink Shared Channel (PUSCH) operation. For DL operation, systems can support cri-RI-CQI reporting and Type II port selection codebook.

(13) However, so far, new radio (NR) has not adequately exploited the channel time domain correlation. Wireless channel is typically time varying, and channel has certain coherence time depending on UE movement speed, and the environment change rate. However, this information is typically not reported by the UE to the network. For Rel-18 MIMO enhancement, there is interest to support reporting of time domain channel properties (TDCP).

(14) However, details regarding such reporting have not been established. It is therefore desirable to specify UE reporting of the time domain channel properties. CSI reporting enhancements for high/medium UE velocities by exploiting time-domain correlation/Doppler-domain information to assist DL precoding, targeting FR1, may include: Rel-16/17 Type-2 codebook refinement, without modification to the spatial and frequency domain basis; and/or UE reporting of time-domain channel properties measured via CSI-RS for tracking.

(15) The measurement of time domain channel properties is based on Tracking Reference Signal (TRS), i.e., “CSI-RS for tracking.” Currently, NR does not support CSI reporting based on TRS. Embodiments herein comprise design details to support UE CSI report of time domain channel properties UE processing related design for reporting time domain channel properties.

Embodiments herein consider Uplink Control Information (UCI) multiplexing, UCI omission, Interference Measurement Resource (IMR) configuration, and phase continuity handling.

(16) FIG. 1 illustrates a signal flow diagram **100** for supporting UE CSI report of time domain

channel properties. The network node **104** may encode and send the UE **102** a CSI report configuration (CSI-ReportConfig **106**). The network node **104** may use the CSI-ReportConfig **106** to configure the CSI report **112**. The UE **102** may decode the CSI-ReportConfig **106** to determine a configuration (e.g., details and timing) for the CSI related reference signal **110** and the CSI report **112**. The CSI-ReportConfig **106** may comprise information related to the CSI related reference signal **110** and the desired information in the CSI report **112**. For instance, the CSI-ReportConfig **106** may include a configuration for the UE **102** to measure and report TDCP based on CSI-RS for TRS.

(17) For aperiodic CSI reports, the network node **104** may transmit a CSI report trigger **108** to the UE **102**. The network node **104** may transmit the CSI related reference signal **110** to the UE **102**, and the UE **102** may perform measurements on the CSI related reference signal **110**. The UE **102** may report the measurements in the CSI report **112**. In some embodiments, the UE **102** may use UCI multiplexing for the CSI report **112**.

(18) There are a significant number of different metrics that the UE **102** may report on. In some embodiments, when UE **102** is configured to report time domain channel properties (TDCP) based on CSI-RS for tracking (TRS), in the same CSI-ReportConfig the UE **102** is not expected to be configured to report other report quantity (reportQuantity). In some embodiments, when UE **102** is configured to report TDCP based on CSI-RS for TRS, in the same CSI-ReportConfig, the UE **102** can be configured to report other report quantity (reportQuantity).

(19) In some embodiments, UCI multiplexing may be used by the UE **102** to report multiple CSIs in the same report. Further, there may be restrictions on what measurements the UE **102** may report. For example, when the UE **102** is configured to report TDCP based on CSI-RS for TRS, in the same CSI-ReportConfig, if UE **102** can be configured to report other report quantity (reportQuantity), the simultaneously reported reportQuantity may be limited to one or multiple of the following choices: layer 1 or physical layer (L1)-Reference Signal Received Power (RSRP); CapabilityIndex; L1-Signal to Interference and Noise Ratio (SINR); and Channel Quality Indicator (CQI). In some embodiments, the simultaneously reported reportQuantity cannot be: CSI-Resource Indicator (CRI), synchronization signal block Resource Indicator (SSBRI), rank indicator (RI), Precoding Matrix Indicator (PMI), and the strongest layer indicator (LI). In other words, the network node **104** may not request the UE **102** to provide CRI, SSBRI, RI, PMI, or LI in the same report as TDCP.

(20) The UE **102** may encode the CSI report **112** in a standardized order so the network node **104** may know how to decode the CSI report **112**. For example, FIG. 2 illustrates a UCI assembly **200** in accordance with some embodiments. When UE is configured to report TDCP based on CSI-RS for TRS, in the same CSI-ReportConfig **106** as other report quantity, the UCI assembly **200** may be formatted as illustrated. As shown, the UCI assembly **200** may include a first field **202** with L1-RSRP/L1-SINR/CQI. After the first field **202**, the UCI assembly **200** may include a capability index field **204**. A third field after the capability index field **204** may comprise the TDCP **206**.

(21) Additionally, the network node **104** may configure the reporting of other report quantities when the UE **102** is configured to report TDCP based on CSI-RS for TRS. The network node **104** may configure the UE **102** to report the other report quantities together with the TDCP via various means. In some embodiments, the network node **104** may configure whether the UE **102** is to report other quantity simultaneously via Radio Resource Control (RRC). In some embodiments, the network node **104** may configure whether the UE **102** is to report other quantity simultaneously via Medium Access Control Control Element (MAC-CE). In some embodiments, at least for aperiodic (AP)-CSI, the network node **104** may indicate whether UE **102** to report other quantity simultaneously via Downlink Control Information (DCI).

(22) The CSI report **112** or UCI may be carried over a physical channel (e.g., PUSCH or physical uplink control channel (PUCCH)). It is possible that the network node **104** may not be able to support all of the bits in the CSI report **112** due to scheduling. Therefore, in some embodiments, the

UE **102** may omit data from the CSI report **112** based on priority.

(23) UCI Omission

(24) UCI omission may be used due to the number of bits scheduled by the network. If the CSI report will exceed a payload size the UE may omit the TDCP and/or other report quantity according to a defined priority. For example, there may not be a sufficient number of bits for all of CSI reporting in the UCI assembly **200**. Accordingly, some embodiments may support UCI omission. When UE is configured to report TDCP based on CSI-RS for tracking (TRS), UCI omission and CSI processing unit (CPU) handling, for the same CSI reporting type (aperiodic or semi-persistent or periodic), the priority of TDCP CSI may be one of the following options.

(25) In some embodiments, UCI omission may be applied using two levels. For example, in a first option, $\text{TDCP CSI} = \text{L1-RSRP/L1-SINR} > \text{other CSI}$. In other words, in some embodiments, the TDCP CSI has equal priority as the L1-RSRP/L1-SINR, and other CSI information is a lower priority. Thus, if the payload is not large enough to support all the information in a CSI report **112** than TDCP CSI and L1-RSRP/L1-SINR will have priority and other CSI may be dropped. In some embodiments, other CSI may include CapabilityIndex and CQI.

(26) A second option may be $\text{L1-RSRP/L1-SINR} > \text{other CSI} = \text{TDCP CSI}$. In other words, in some embodiments, the L1-RSRP/L1-SINR has a higher priority, and TDCP CSI and other CSI information have a lower priority. Thus, in some embodiments, if the payload is not large enough to support all the information in a CSI report **112** than L1-RSRP/L1-SINR will have priority, and TDCP CSI and other CSI may be dropped.

(27) In some embodiments, UCI omission may be applied using three levels. Using the two level approach may be less complex to implement, whereas using the two level approach may allow a greater ability to define a desired priority. As a third UCI priority option, $\text{TDCP CSI} > \text{L1-RSRP/L1-SINR} > \text{other CSI}$. As a fourth priority option, $\text{L1-RSRP/L1-SINR} > \text{TDCP CSI} > \text{other CSI}$, and as a fifth priority option, $\text{L1-RSRP/L1-SINR} > \text{other CSI} > \text{TDCP CSI}$. Different embodiments may employ different priorities as desired.

(28) In some embodiments, when UE **102** is configured to report TDCP based on CSI-RS for tracking (TRS) in a same CSI-ReportConfig with other report quantity (reportQuantity) simultaneously, the UE **102** may perform one of the following options. In a first option, if the UE **102** omits/drops CSI, the UE **102** drops all the report quantity in the same CSI-ReportConfig. In a second option (option 2), if the UE **102** omits/drops CSI, the UE **102** drops different parts of report quantity in the same CSI-ReportConfig with different priority. For example, in sub option (option 2.1), the UE **102** may first drop other report quantity, then the UE **102** drops TDCP if needed. As another sub option (option 2.2) the UE **102** first drops TDCP, then, UE drops other report quantity if needed.

(29) In some embodiments, discontinuous reception (DRX) may be used to reduce power consumption. For example, the UE **102** may listen for information from the network node **104** periodically based on a wake up signal from the network node **104**. In some embodiments, when the UE **102** is configured to report TDCP based on CSI-RS for tracking (TRS), when drx-onDurationTimer in DRX-Config is not started the UE may perform one of the following options. In a first option, the UE **102** is not required to measure TDCP CSI. In a second option, the UE **102** may still measure TDCP CSI with the following restrictions.

(30) For option 2, the UE **102** may be required to measure only the periodic TDCP CSI. Further, the UE **102** may report, to the network node **104**, the UE **102** capability of to support reporting TDCP CSI measured when drx-onDurationTimer in DRX-Config is not started. In some embodiments, the network node **104** may explicitly configure whether the UE **102** shall report TDCP CSI measured when drx-onDurationTimer in DRX-Config is not started. In some embodiments, a new RRC configuration may be introduced for the configuration. The network node **104** may send the new RRC configuration to the UE **102** to configure whether the UE **102** should report the TDCP CSI. In some embodiments, an existing RRC configuration (e.g., either ps-

TransmitPeriodicCSI or ps-TransmitPeriodicL1-RSRP) may be reused for the configuration. (31) CSI may include two types of measurement resources, CMR and IMR. CMR may be used to measure a channel (e.g., measure the TDCP). IMR may be used to measure the strength of interference from other resources and signals not intended for a UE. IMR measurements may include L1-SINR and CQI.

(32) IMR Configuration

(33) In some embodiments, when UE is configured to report time domain channel properties (TDCP) based on CSI-RS for tracking (TRS), the following are the options regarding IMR configuration.

(34) In a first option, the IMR cannot be configured, neither non-zero power (NZP)-IMR (NZP-CSI-RS) nor zero power (ZP)-IMR (e.g., CSI-IM). In other words, the network node **104** may not be able to configure the UE **102** to measure and report IMR.

(35) In a second option, the network node **104** may configure the UE **102** to measure and report the IMR with some restrictions. The CSI-ReportConfig **106** may include configuration details for both IMR and CMR, the UE **102** may measure the IMR and the CMR, and the CSI report **112** may include both measurements from IMR and CMR. In some embodiments, only ZP-IMR (e.g., CSI-IM) can be configured. In some embodiments, either ZP-IMR (e.g., CSI-IM) or NZP-IMR (NZP-CSI-RS) can be configured, but not both in the same CSI-ReportConfig. In some embodiments, ZP-IMR or NZP-IMR or ZP-IMR+NZP-IMR can be configured.

(36) In some embodiments, when UE is configured to report TDCP based on CSI-RS for tracking (TRS), if IMR can be configured, the following are two options regarding IMR configuration. In option 1, the IMR resource can be configured one to one mapped to the configured CSI-RS (CMR). For example, if TRS contains four CSI-RS as CMR, the network configures four IMR. Each IMR is associated with its corresponding CMR, assuming the same quasi co-location (QCL). In option 2, only one IMR resource can be configured. The one IMR may be associated with multiple CMR. For example, if TRS contains four CSI-RS as CMR, and the network configures one IMR, the one IMR may correspond to the four CMR.

(37) The scheduling of the IMR may be configured to reduce power consumption at the UE **102**. For example, when UE **102** is configured to report time domain channel properties (TDCP) based on CSI-RS for tracking (TRS), if IMR can be configured, scheduling restrictions for IMR and CMR may be implemented. FIG. 3 and FIG. 4 illustrate two potential scheduling restrictions.

(38) FIG. 3 illustrates a scheduling restriction **300** based on the DRX active time **302**. The IMR and CMR may be configured in the same DRX active time **302** as shown in FIG. 3. This scheduling restriction **300** may ensure that the UE does not perform CMR and IMR activities occur outside of the DRX active time **302**. In the illustrated embodiment, the DRX active time **302** includes four slots. The first two IMRs **304** occur in a first slot of the DRX active time **302** and the second two IMRs **306** occur in a last slot of the DRX active time **302**. As shown, the CMRs **308** corresponding to the IMRs may occur during the second and third slots. Accordingly, the UE may perform multiple CMR measurements and associated IMR measurements in the same DRX active time **302**.

(39) FIG. 4 illustrates a scheduling restriction **400** based on a number of slots. The IMR and CMR may be configured within X slots (e.g., X=2, 3, or 4). In the illustrated embodiment, the IMR and CMR are configured within two slots **402**. If there is no restriction, the IMR and CMR measurements may occur across an extended period of time. During that time, the UE may be consuming power and storing a measurement while waiting for additional measurements before sending the report. Limiting the number of slots able to be configured for IMR and CMR may reduce UE power consumption and UE memory usage. In some embodiments, the IMR and CMR may be restricted to be both configured in the same DRX active time and within X slots.

(40) Phase Continuity Handling

(41) TDCP measurements may be based on the channel between the UE and the network node. For instance, the UE may take multiple measurements of the channel and determine phase change at the

different locations. The UE can use the measurements to derive phase change and Doppler shift. However, the difference in the measurements may be skewed if a phase change is introduced between the measurements. For instance, if the UE switches between receiving and transmitting (e.g., a duplex direction change) between the measurements, the phase of the channel may be interrupted for the channel. Accordingly, the difference between measurements may be inaccurate due to the phase interruption leading to an inaccurate TDCP measurement.

(42) FIG. 5 illustrates a method **500** for a UE to handle phase continuity. The UE may receive **502**, from a network node, a CSI-ReportConfig. The CSI-ReportConfig may include details regarding measurement and reporting of TDCP based on CSI-RS resources for TRS. However, the UE may execute **504** a duplex direction change which may interfere with the TDRP measurement. The duplex direction change may occur when the UE switches from receiving to transmitting and back to receiving.

(43) The UE may determine **506** invalid CSI-RS resources and valid CSI-RS resources based on a timing of the duplex direction change. In some embodiments, when the UE is configured to report TDCP based on CSI-RS for TRS, UE may measure TRS only when UE can ensure the phase continuity during the TRS measurement. The phase continuity is disrupted when the UE performs duplex direction change between CSI-RS resource in the same TRS set. In some embodiments, when the phase continuity is disrupted, the UE is not required to measure the TRS transmission in which the phase continuity is disrupted. The UE may measure **508** the TDCP based on the valid CSI-RS resources, and report, to the network node, the TDCP based on the valid CSI-RS resources via a CSI report.

(44) In some embodiments, the determination of valid and invalid resources may depend on the timing of the CSI-RS resources relative to the duplex direction change. For example, FIG. 6 illustrates a timeline **600** of a phase continuity interruption according to a first case. The timeline **600** includes four CSI-RS resources in a TRS set, and a duplex direction change **602**. The illustrated timeline **600** is referred to herein as case **1**. In case **1**, the phase continuity interruption happens between two CSI-RS resources in the same slot.

(45) In some embodiments, for interruptions as shown in the case **1** (i.e., phase continuity interruption happens between two CSI-RS resources in the same slot), the following options may be employed. In some embodiments, the UE may consider all CSI-RS resources invalid for TDCP measurement, including CSI-RS 1, CSI-RS 2, CSI-RS 3, and CSI-RS 4. The UE may not measure these invalid CSI RS resources.

(46) In some embodiments, a UE may consider some CSI-RS resources still valid for TDCP measurement even with the duplex direction change **602**. In some embodiments, the UE may determine that valid CSI-RS resources include CSI-RS 3 and CSI-RS 4 that may be used for TDCP measurements. In some embodiments, the UE may consider CSI-RS 2, CSI-RS3, and CSI-RS 4 valid CSI-RS resources that may be used for TDCP measurements. As CSI-RS 2 is in a different slot than CSI-RS3, and CSI-RS 4, the UE may perform additional time domain processing to account for a different amount of time between CSI-RS 2 and CSI-RS3 than is between CSI-RS3 and CSI-RS 4. The additional use of CSI-RS 2 may provide a more accurate TDCP measurement at the cost of potential additional processing.

(47) FIG. 7 illustrates a timeline **700** of a second case of phase continuity interruption. For interruption case **2**, the phase continuity interruption (e.g., duplex direction change **702**) happens between two pairs of CSI-RS resources in adjacent slot. In some embodiments, the UE may determine that all CSI-RS resources are invalid for TDCP measurement, including CSI-RS 1, CSI-RS 2, CSI-RS 3, and CSI-RS 4. In some embodiments, the UE may determine that all CSI-RS resources are valid for TDCP measurement, including CSI-RS 1, CSI-RS 2, CSI-RS 3, and CSI-RS 4. Because the duplex direction change **702** occurs between pairs of CSI-RS resources, the UE may measure TDCP based on CSI-RS resources in the first slot and then measure the TDCP based on the CSI-RS resources of the second slot and combine those two TDCP measurements to suppress

measurement noise and obtain a more accurate measurement. In some embodiments, the UE may determine that only one pair of CSI-RS resources are valid for TDCP measurement (e.g., either {CSI-RS 1, CSI-RS 2} or {CSI-RS 3, CSI-RS 4}).

(48) In some embodiments, the UE's response to the first case and the second case as described with reference to FIG. 7 and FIG. 8 may be performed by a same embodiment. That is, the UE may behave differently depending on where the duplex direction change **702** occurs.

(49) In some embodiments, when the UE is configured to report TDCP based on CSI-RS for TRS and when UE is not required to measure the TRS transmission in which the phase continuity is disrupted, the following embodiments provide details regarding how the invalid CSI-RS resources may be counted for CSI processing unit (CPU) occupation and active CSI-RS. In some embodiments, the invalid CSI-RS that cannot be used for TDCP measurement may still occupy the CPU. In some embodiments, for CPU occupation, invalid an CSI-RS does not occupy the CPU. In some embodiments, for active CSI-RS, an invalid CSI-RS is still counted as active CSI-RS. In some embodiments, for active CSI-RS, an invalid CSI-RS is not counted as active CSI-RS.

(50) FIG. 8 illustrates an example architecture of a wireless communication system **800**, according to embodiments disclosed herein. The following description is provided for an example wireless communication system **800** that operates in conjunction with the LTE system standards and/or 5G or NR system standards as provided by 3GPP technical specifications.

(51) As shown by FIG. 8, the wireless communication system **800** includes UE **802** and UE **804** (although any number of UEs may be used). In this example, the UE **802** and the UE **804** are illustrated as smartphones (e.g., handheld touchscreen mobile computing devices connectable to one or more cellular networks), but may also comprise any mobile or non-mobile computing device configured for wireless communication.

(52) The UE **802** and UE **804** may be configured to communicatively couple with a RAN **806**. In embodiments, the RAN **806** may be NG-RAN, E-UTRAN, etc. The UE **802** and UE **804** utilize connections (or channels) (shown as connection **808** and connection **810**, respectively) with the RAN **806**, each of which comprises a physical communications interface. The RAN **806** can include one or more base stations (such as base station **812** and base station **814**) that enable the connection **808** and connection **810**.

(53) In this example, the connection **808** and connection **810** are air interfaces to enable such communicative coupling, and may be consistent with RAT(s) used by the RAN **806**, such as, for example, an LTE and/or NR.

(54) In some embodiments, the UE **802** and UE **804** may also directly exchange communication data via a sidelink interface **816**. The UE **804** is shown to be configured to access an access point (shown as AP **818**) via connection **820**. By way of example, the connection **820** can comprise a local wireless connection, such as a connection consistent with any IEEE 802.11 protocol, wherein the AP **818** may comprise a Wi-Fi® router. In this example, the AP **818** may be connected to another network (for example, the Internet) without going through a CN **824**.

(55) In embodiments, the UE **802** and UE **804** can be configured to communicate using orthogonal frequency division multiplexing (OFDM) communication signals with each other or with the base station **812** and/or the base station **814** over a multicarrier communication channel in accordance with various communication techniques, such as, but not limited to, an orthogonal frequency division multiple access (OFDMA) communication technique (e.g., for downlink communications) or a single carrier frequency division multiple access (SC-FDMA) communication technique (e.g., for uplink and ProSe or sidelink communications), although the scope of the embodiments is not limited in this respect. The OFDM signals can comprise a plurality of orthogonal subcarriers.

(56) In some embodiments, all or parts of the base station **812** or base station **814** may be implemented as one or more software entities running on server computers as part of a virtual network. In addition, or in other embodiments, the base station **812** or base station **814** may be configured to communicate with one another via interface **822**. In embodiments where the wireless

communication system **800** is an LTE system (e.g., when the CN **824** is an EPC), the interface **822** may be an X2 interface. The X2 interface may be defined between two or more base stations (e.g., two or more eNBs and the like) that connect to an EPC, and/or between two eNBs connecting to the EPC. In embodiments where the wireless communication system **800** is an NR system (e.g., when CN **824** is a 5GC), the interface **822** may be an Xn interface. The Xn interface is defined between two or more base stations (e.g., two or more gNBs and the like) that connect to 5GC, between a base station **812** (e.g., a gNB) connecting to 5GC and an eNB, and/or between two eNBs connecting to 5GC (e.g., CN **824**).

(57) The RAN **806** is shown to be communicatively coupled to the CN **824**. The CN **824** may comprise one or more network elements **826**, which are configured to offer various data and telecommunications services to customers/subscribers (e.g., users of UE **802** and UE **804**) who are connected to the CN **824** via the RAN **806**. The components of the CN **824** may be implemented in one physical device or separate physical devices including components to read and execute instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium).

(58) In embodiments, the CN **824** may be an EPC, and the RAN **806** may be connected with the CN **824** via an S1 interface **828**. In embodiments, the S1 interface **828** may be split into two parts, an S1 user plane (S1-U) interface, which carries traffic data between the base station **812** or base station **814** and a serving gateway (S-GW), and the S1-MME interface, which is a signaling interface between the base station **812** or base station **814** and mobility management entities (MMEs).

(59) In embodiments, the CN **824** may be a 5GC, and the RAN **806** may be connected with the CN **824** via an NG interface **828**. In embodiments, the NG interface **828** may be split into two parts, an NG user plane (NG-U) interface, which carries traffic data between the base station **812** or base station **814** and a user plane function (UPF), and the S1 control plane (NG-C) interface, which is a signaling interface between the base station **812** or base station **814** and access and mobility management functions (AMFs).

(60) Generally, an application server **830** may be an element offering applications that use internet protocol (IP) bearer resources with the CN **824** (e.g., packet switched data services). The application server **830** can also be configured to support one or more communication services (e.g., VoIP sessions, group communication sessions, etc.) for the UE **802** and UE **804** via the CN **824**. The application server **830** may communicate with the CN **824** through an IP communications interface **832**.

(61) FIG. 9 illustrates a system **900** for performing signaling **934** between a wireless device **902** and a network device **918**, according to embodiments disclosed herein. The system **900** may be a portion of a wireless communications system as herein described. The wireless device **902** may be, for example, a UE of a wireless communication system. The network device **918** may be, for example, a base station (e.g., an eNB or a gNB) of a wireless communication system.

(62) The wireless device **902** may include one or more processor(s) **904**. The processor(s) **904** may execute instructions such that various operations of the wireless device **902** are performed, as described herein. The processor(s) **904** may include one or more baseband processors implemented using, for example, a central processing unit (CPU), a digital signal processor (DSP), an application specific integrated circuit (ASIC), a controller, a field programmable gate array (FPGA) device, another hardware device, a firmware device, or any combination thereof configured to perform the operations described herein.

(63) The wireless device **902** may include a memory **906**. The memory **906** may be a non-transitory computer-readable storage medium that stores instructions **908** (which may include, for example, the instructions being executed by the processor(s) **904**). The instructions **908** may also be referred to as program code or a computer program. The memory **906** may also store data used by, and results computed by, the processor(s) **904**.

(64) The wireless device **902** may include one or more transceiver(s) **910** that may include radio frequency (RF) transmitter and/or receiver circuitry that use the antenna(s) **912** of the wireless device **902** to facilitate signaling (e.g., the signaling **934**) to and/or from the wireless device **902** with other devices (e.g., the network device **918**) according to corresponding RATs.

(65) The wireless device **902** may include one or more antenna(s) **912** (e.g., one, two, four, or more). For embodiments with multiple antenna(s) **912**, the wireless device **902** may leverage the spatial diversity of such multiple antenna(s) **912** to send and/or receive multiple different data streams on the same time and frequency resources. This behavior may be referred to as, for example, multiple input multiple output (MIMO) behavior (referring to the multiple antennas used at each of a transmitting device and a receiving device that enable this aspect). MIMO transmissions by the wireless device **902** may be accomplished according to precoding (or digital beamforming) that is applied at the wireless device **902** that multiplexes the data streams across the antenna(s) **912** according to known or assumed channel characteristics such that each data stream is received with an appropriate signal strength relative to other streams and at a desired location in the spatial domain (e.g., the location of a receiver associated with that data stream). Certain embodiments may use single user MIMO (SU-MIMO) methods (where the data streams are all directed to a single receiver) and/or multi user MIMO (MU-MIMO) methods (where individual data streams may be directed to individual (different) receivers in different locations in the spatial domain).

(66) In certain embodiments having multiple antennas, the wireless device **902** may implement analog beamforming techniques, whereby phases of the signals sent by the antenna(s) **912** are relatively adjusted such that the (joint) transmission of the antenna(s) **912** can be directed (this is sometimes referred to as beam steering).

(67) The wireless device **902** may include one or more interface(s) **914**. The interface(s) **914** may be used to provide input to or output from the wireless device **902**. For example, a wireless device **902** that is a UE may include interface(s) **914** such as microphones, speakers, a touchscreen, buttons, and the like in order to allow for input and/or output to the UE by a user of the UE. Other interfaces of such a UE may be made up of transmitters, receivers, and other circuitry (e.g., other than the transceiver(s) **910**/antenna(s) **912** already described) that allow for communication between the UE and other devices and may operate according to known protocols (e.g., Wi-Fi®, Bluetooth®, and the like).

(68) The wireless device **902** may include a TDCP reporting module **916**. The TDCP reporting module **916** may be implemented via hardware, software, or combinations thereof. For example, the TDCP reporting module **916** may be implemented as a processor, circuit, and/or instructions **908** stored in the memory **906** and executed by the processor(s) **904**. In some examples, the TDCP reporting module **916** may be integrated within the processor(s) **904** and/or the transceiver(s) **910**. For example, the TDCP reporting module **916** may be implemented by a combination of software components (e.g., executed by a DSP or a general processor) and hardware components (e.g., logic gates and circuitry) within the processor(s) **904** or the transceiver(s) **910**.

(69) The TDCP reporting module **916** may be used for various aspects of the present disclosure, for example, aspects of FIGS. **1-5**. The TDCP reporting module **916** is configured to report CSI of time domain channel properties.

(70) The network device **918** may include one or more processor(s) **920**. The processor(s) **920** may execute instructions such that various operations of the network device **918** are performed, as described herein. The processor(s) **920** may include one or more baseband processors implemented using, for example, a CPU, a DSP, an ASIC, a controller, an FPGA device, another hardware device, a firmware device, or any combination thereof configured to perform the operations described herein.

(71) The network device **918** may include a memory **922**. The memory **922** may be a non-transitory computer-readable storage medium that stores instructions **924** (which may include, for example,

the instructions being executed by the processor(s) **920**). The instructions **924** may also be referred to as program code or a computer program. The memory **922** may also store data used by, and results computed by, the processor(s) **920**.

(72) The network device **918** may include one or more transceiver(s) **926** that may include RF transmitter and/or receiver circuitry that use the antenna(s) **228** of the network device **918** to facilitate signaling (e.g., the signaling **934**) to and/or from the network device **918** with other devices (e.g., the wireless device **902**) according to corresponding RATs.

(73) The network device **918** may include one or more antenna(s) **928** (e.g., one, two, four, or more). In embodiments having multiple antenna(s) **928**, the network device **918** may perform MIMO, digital beamforming, analog beamforming, beam steering, etc., as has been described.

(74) The network device **918** may include one or more interface(s) **930**. The interface(s) **930** may be used to provide input to or output from the network device **918**. For example, a network device **918** that is a base station may include interface(s) **930** made up of transmitters, receivers, and other circuitry (e.g., other than the transceiver(s) **926**/antenna(s) **928** already described) that enables the base station to communicate with other equipment in a core network, and/or that enables the base station to communicate with external networks, computers, databases, and the like for purposes of operations, administration, and maintenance of the base station or other equipment operably connected thereto.

(75) The network device **918** may include a TDCP reporting module **932**. The TDCP reporting module **932** may be implemented via hardware, software, or combinations thereof. For example, the TDCP reporting module **932** may be implemented as a processor, circuit, and/or instructions **924** stored in the memory **922** and executed by the processor(s) **920**. In some examples, the TDCP reporting module **932** may be integrated within the processor(s) **920** and/or the transceiver(s) **926**. For example, the TDCP reporting module **932** may be implemented by a combination of software components (e.g., executed by a DSP or a general processor) and hardware components (e.g., logic gates and circuitry) within the processor(s) **920** or the transceiver(s) **926**.

(76) The TDCP reporting module **932** may be used for various aspects of the present disclosure, for example, aspects of FIGS. **1-5**. The TDCP reporting module **932** is configured to support UE CSI report of time domain channel properties.

(77) Embodiments contemplated herein include an apparatus comprising means to perform one or more elements of the methods disclosed herein. This apparatus may be, for example, an apparatus of a UE (such as a wireless device **902** that is a UE, as described herein).

(78) Embodiments contemplated herein include one or more non-transitory computer-readable media comprising instructions to cause an electronic device, upon execution of the instructions by one or more processors of the electronic device, to perform one or more elements of the methods disclosed herein. This non-transitory computer-readable media may be, for example, a memory of a UE (such as a memory **906** of a wireless device **902** that is a UE, as described herein).

(79) Embodiments contemplated herein include an apparatus comprising logic, modules, or circuitry to perform one or more elements of the methods disclosed herein. This apparatus may be, for example, an apparatus of a UE (such as a wireless device **902** that is a UE, as described herein).

(80) Embodiments contemplated herein include an apparatus comprising: one or more processors and one or more computer-readable media comprising instructions that, when executed by the one or more processors, cause the one or more processors to perform one or more elements of the methods disclosed herein. This apparatus may be, for example, an apparatus of a UE (such as a wireless device **902** that is a UE, as described herein).

(81) Embodiments contemplated herein include a signal as described in or related to one or more elements of the methods disclosed herein.

(82) Embodiments contemplated herein include a computer program or computer program product comprising instructions, wherein execution of the program by a processor is to cause the processor to carry out one or more elements of the methods disclosed herein. The processor may be a

processor of a UE (such as a processor(s) **904** of a wireless device **902** that is a UE, as described herein). These instructions may be, for example, located in the processor and/or on a memory of the UE (such as a memory **906** of a wireless device **902** that is a UE, as described herein).

(83) Embodiments contemplated herein include an apparatus comprising means to perform one or more elements of the methods disclosed herein. This apparatus may be, for example, an apparatus of a base station (such as a network device **918** that is a base station, as described herein).

(84) Embodiments contemplated herein include one or more non-transitory computer-readable media comprising instructions to cause an electronic device, upon execution of the instructions by one or more processors of the electronic device, to perform one or more elements of the methods disclosed herein. This non-transitory computer-readable media may be, for example, a memory of a base station (such as a memory **922** of a network device **918** that is a base station, as described herein).

(85) Embodiments contemplated herein include an apparatus comprising logic, modules, or circuitry to perform one or more elements of the methods disclosed herein. This apparatus may be, for example, an apparatus of a base station (such as a network device **918** that is a base station, as described herein).

(86) Embodiments contemplated herein include an apparatus comprising: one or more processors and one or more computer-readable media comprising instructions that, when executed by the one or more processors, cause the one or more processors to perform one or more elements of the methods disclosed herein. This apparatus may be, for example, an apparatus of a base station (such as a network device **918** that is a base station, as described herein).

(87) Embodiments contemplated herein include a signal as described in or related to one or more elements of the methods disclosed herein.

(88) Embodiments contemplated herein include a computer program or computer program product comprising instructions, wherein execution of the program by a processing element is to cause the processing element to carry out one or more elements of the methods disclosed herein. The processor may be a processor of a base station (such as a processor(s) **920** of a network device **918** that is a base station, as described herein). These instructions may be, for example, located in the processor and/or on a memory of the base station (such as a memory **922** of a network device **918** that is a base station, as described herein).

(89) For one or more embodiments, at least one of the components set forth in one or more of the preceding figures may be configured to perform one or more operations, techniques, processes, and/or methods as set forth herein. For example, a baseband processor as described herein in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth herein. For another example, circuitry associated with a UE, base station, network element, etc. as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth herein.

(90) Any of the above described embodiments may be combined with any other embodiment (or combination of embodiments), unless explicitly stated otherwise. The foregoing description of one or more implementations provides illustration and description, but is not intended to be exhaustive or to limit the scope of embodiments to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of various embodiments.

(91) Embodiments and implementations of the systems and methods described herein may include various operations, which may be embodied in machine-executable instructions to be executed by a computer system. A computer system may include one or more general-purpose or special-purpose computers (or other electronic devices). The computer system may include hardware components that include specific logic for performing the operations or may include a combination of hardware, software, and/or firmware.

(92) It should be recognized that the systems described herein include descriptions of specific embodiments. These embodiments can be combined into single systems, partially combined into other systems, split into multiple systems or divided or combined in other ways. In addition, it is contemplated that parameters, attributes, aspects, etc. of one embodiment can be used in another embodiment. The parameters, attributes, aspects, etc. are merely described in one or more embodiments for clarity, and it is recognized that the parameters, attributes, aspects, etc. can be combined with or substituted for parameters, attributes, aspects, etc. of another embodiment unless specifically disclaimed herein.

(93) It is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.

(94) Although the foregoing has been described in some detail for purposes of clarity, it will be apparent that certain changes and modifications may be made without departing from the principles thereof. It should be noted that there are many alternative ways of implementing both the processes and apparatuses described herein. Accordingly, the present embodiments are to be considered illustrative and not restrictive, and the description is not to be limited to the details given herein, but may be modified within the scope and equivalents of the appended claims.

Claims

1. A method for a user equipment (UE), the method comprising: receiving, from a network node, a channel state information (CSI) report configuration (CSI-ReportConfig), the CSI-ReportConfig comprising details regarding measurement and reporting of time domain channel properties (TDCP) based on a CSI reference signal (CSI-RS) for tracking reference signal (TRS); executing a duplex direction change; determining invalid CSI-RS resources and valid CSI-RS resources based on a timing of the duplex direction change, wherein the invalid CSI-RS resources occupy a CSI processing unit (CPU) and are counted as active CSI-RS; measuring the TDCP based on the valid CSI-RS resources; and reporting, to the network node, the TDCP based on the valid CSI-RS resources via a CSI report.
2. The method of claim 1, wherein the UE does not measure the invalid CSI-RS resources.
3. The method of claim 1, further comprising determining a location of the duplex change relative to pairs of CSI-RS resources.
4. The method of claim 3 wherein when the duplex change is between a first pair of CSI-RS resources, the first pair is invalid and any remaining pairs of CSI-RS resources are valid.
5. The method of claim 3, wherein when the duplex change is between a first pair of CSI-RS resources, one CSI-RS resource from the first pair is invalid and an other CSI-RS resource from the first pair is grouped with remaining pairs of CSI-RS resources, wherein the other CSI-RS resource from the first pair and the remaining pairs are valid.
6. The method of claim 3, wherein when the duplex change is between two pairs of CSI-RS resources, all CSI-RS resources are valid.
7. The method of claim 3, wherein when the duplex change is between two pairs of CSI-RS resources, all CSI-RS resources are invalid.
8. The method of claim 3, wherein when the duplex change is between two pairs of CSI-RS resources, the UE selects one of the two pairs for TDCP measurement.
9. The method of claim 1, wherein the invalid CSI-RS resources do not occupy a CSI processing unit (CPU) and are not counted as active CSI-RS.
10. A user equipment (UE) comprising: a processor; and a memory storing instructions that, when

executed by the processor, configure the UE to: receive, from a network node, a channel state information (CSI) report configuration (CSI-ReportConfig), the CSI-ReportConfig comprising details regarding measurement and reporting of time domain channel properties (TDCP) based on a CSI reference signal (CSI-RS) for tracking reference signal (TRS); execute a duplex direction change; determine invalid CSI-RS resources and valid CSI-RS resources based on a timing of the duplex direction change, wherein the invalid CSI-RS resources do not occupy a CSI process unit (CPU) and are not counted as active CSI-RS; measure the TDCP based on the valid CSI-RS resources; and report, to the network node, the TDCP based on the valid CSI-RS resources via a CSI report.

11. The UE of claim 10, wherein the UE does not measure the invalid CSI-RS resources.

12. The UE of claim 10, wherein the instructions further configure the UE to determine a location of the duplex change relative to pairs of CSI-RS resources.

13. The UE of claim 12, wherein when the duplex change is between a first pair of CSI-RS resources, the first pair is invalid and any remain pairs of CSI-RS resources are valid.

14. The UE of claim 12, wherein when the duplex change is between a first pair of CSI-RS resources, one CSI-RS resource from the first pair is invalid and an other CSI-RS resource from the first pair is grouped with remaining pairs of CSI-RS resources, wherein the other CSI-RS resource from the first pair and the remaining pairs of CSI-RS resources are valid.

15. The UE of claim 12, wherein when the duplex change is between two pairs of CSI-RS resources, all CSI-RS resources are valid.

16. The UE of claim 12, wherein when the duplex change is between two pairs of CSI-RS resources, all CSI-RS resources are invalid.

17. The UE of claim 12, wherein when the duplex change is between two pairs of CSI-RS resources, the UE selects one of the two pairs for TDCP measurement.

18. The UE of claim 10, wherein the invalid CSI-RS resources occupy a CSI process unit (CPU) and are counted as active CSI-RS.

19. A method for a user equipment (UE), the method comprising: receiving, from a network node, a channel state information (CSI) report configuration (CSI-ReportConfig), the CSI-ReportConfig comprising details regarding measurement and reporting of time domain channel properties (TDCP) based on a CSI reference signal (CSI-RS) for tracking reference signal (TRS); executing a duplex direction change; determining invalid CSI-RS resources and valid CSI-RS resources based on a timing of the duplex direction change; determining a location of the duplex change relative to pairs of CSI-RS resources, wherein when the duplex change is between a first pair of CSI-RS resources, one CSI-RS resource from the first pair is invalid and an other CSI-RS resource from the first pair is grouped with remaining pairs of CSI-RS resources, wherein the other CSI-RS resource from the first pair and the remaining pairs are valid; measuring the TDCP based on the valid CSI-RS resources; and reporting, to the network node, the TDCP based on the valid CSI-RS resources via a CSI report.
